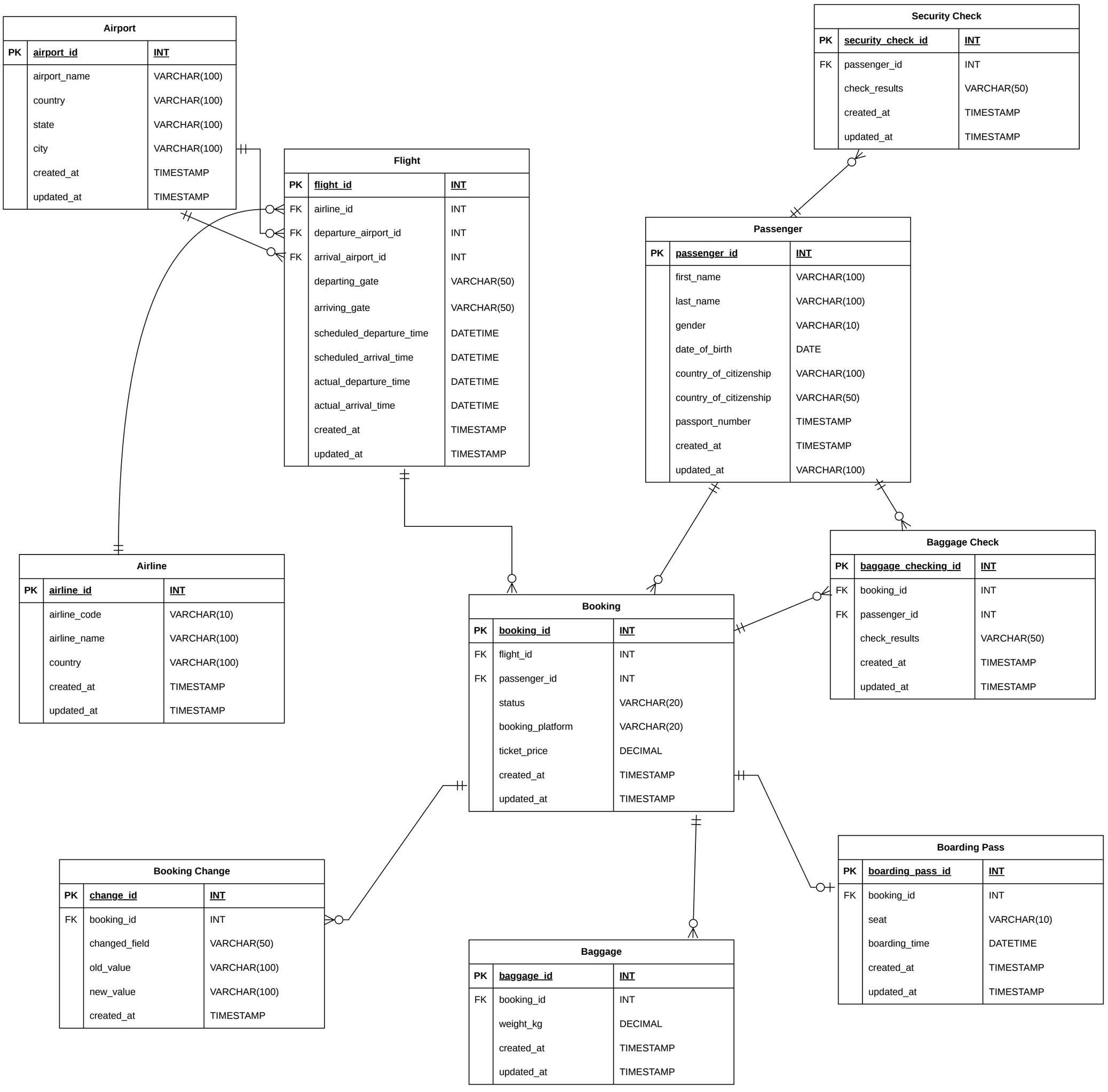

Airport
PK airport_id INT
airport_name VARCHAR(100)
country VARCHAR(100)
state VARCHAR(100)
city VARCHAR(100)
created_at TIMESTAMP
updated_at TIMESTAMP
Flight
PK flight_id INT
FK airline_id INT
FK departure_airport_id INT
FK arrival_airport_id INT
departing_gate VARCHAR(50)
arriving_gate VARCHAR(50)
scheduled_departure_time DATETIME
scheduled_arrival_time DATETIME
actual_departure_time DATETIME
actual_arrival_time DATETIME
created_at TIMESTAMP
updated_at TIMESTAMP
Security Check
PK security_check_id INT
FK passenger_id INT
check_results VARCHAR(50)
created_at TIMESTAMP
updated_at TIMESTAMP
Passenger
PK passenger_id INT
first_name VARCHAR(100)
last_name VARCHAR(100)
gender VARCHAR(10)
date_of_birth DATE
country_of_citizenship VARCHAR(100)
country_of_citizenship VARCHAR(50)
passport_number TIMESTAMP
created_at TIMESTAMP
updated_at VARCHAR(100)
Airline
PK airline_id INT
airline_code VARCHAR(10)
airline_name VARCHAR(100)
country VARCHAR(100)
created_at TIMESTAMP
updated_at TIMESTAMP
Booking
PK booking_id INT
FK flight_id INT
FK passenger_id INT
status VARCHAR(20)
booking_platform VARCHAR(20)
ticket_price DECIMAL
created_at TIMESTAMP
updated_at TIMESTAMP
Baggage Check
PK baggage_checking_id INT
FK booking_id INT
FK passenger_id INT
check_results VARCHAR(50)
created_at TIMESTAMP
updated_at TIMESTAMP
Booking Change
PK change_id INT
FK booking_id INT
changed_field VARCHAR(50)
old_value VARCHAR(100)
new_value VARCHAR(100)
created_at TIMESTAMP
Baggage
PK baggage_id INT
FK booking_id INT
weight_kg DECIMAL
created_at TIMESTAMP
updated_at TIMESTAMP
Boarding Pass
PK boarding_pass_id INT
FK booking_id INT
seat VARCHAR(10)
boarding_time DATETIME
created_at TIMESTAMP
updated_at TIMESTAMP

Entities:

- **Airport** - stores information about each airport.
- **Airline** - stores details of airline companies operating flights.
- **Flight** - records scheduled and actual flight details.
- **Passenger** - personal informationof passengers.
- **Booking** - passenger's reservation on a specific flight.
- **BookingChange** - records the history ofany modifications made to a booking.
- **BoardingPass** - stores the boarding pass issued for a booking.
- **Baggage** - baggage details registered for a booking.
- **BaggageCheck** - records the results of baggage screening for a booking and passenger.
- **SecurityCheck** - records the results of airport security screening for a passenger.

Relationships:

Almost all relationships are optional-one-to-mandatory-many: the "one" side can exist in the database without any related records yet, while the "many" side must always be linked to exactly one parent record to exist at all.

Except Booking ↔BoardingPass, which has one-to-one relationship: each booking can optionally have at most one boarding pass, while each boarding pass belongs to exactly one booking.

Notation key:

- **PK** — Primary Key
- **FK** — Foreign Key